

Week 5 Worksheet 2 Solutions

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July 22, 2026

Exercise 1. Denote the eigenspinors of $S_z = \frac{\hbar}{2}\sigma_z$ for a spin-1/2 particle by $|+\rangle$ and $|-\rangle$, with eigenvalues $\hbar/2$ and $-\hbar/2$, respectively.

a) Write down $S_y = \frac{\hbar}{2}\sigma_y$ and $S_x = \frac{\hbar}{2}\sigma_x$ as

$$a|+\rangle\langle+| + b|+\rangle\langle-| + c|-\rangle\langle+| + d|-\rangle\langle-|;$$

that is, determine a, b, c, d .

b) Explain how you could extract a, b, c, d for S_i by taking $\langle\pm|S_i|\pm\rangle$ or $\langle\pm|S_i|\mp\rangle$.

a) For S_y , $a = 0$, $d = 0$, $b = -i\hbar/2$, and $c = i\hbar/2$. For S_x , $a = d = 0$, $b = \hbar/2$, and $c = \hbar/2$.

b) We could extract a by taking $\langle+|S_i|+\rangle$, b by taking $\langle+|S_i|-\rangle$, c by taking $\langle-|S_i|+\rangle$, and d by taking $\langle-|S_i|-\rangle$.

Exercise 2. Adapted from Griffiths 4.33 Suppose we instead wanted to compute spin for a spin-1/2 particle along an arbitrary direction $\hat{n}$ (the hats in this problem denote “unit vector,” not “operator”).

a) Convince yourself that we can place the z -axis along $\hat{n}$ by rotating first around $\hat{y}$ by an angle β and then around $\hat{z}$ (the old z -axis, not the new one!) by an angle α .

Hint: Write $\hat{n}$ in terms of its polar and azimuthal angles. How are these related to α and β ?

b) Use this to compute the eigenspinor of S_n with eigenvalue $\hbar/2$, $|\hat{n}; +\rangle$: It is given by

$$|\hat{n}; +\rangle = e^{-\frac{i}{2}\alpha\sigma_z} e^{-\frac{i}{2}\beta\sigma_y} |+\rangle,$$

where $|+\rangle$ is the eigenvector of S_z with eigenvalue $\hbar/2$.

Hint: You can use the formula which you will prove on the review:

$$e^{-i\theta\hat{n}\cdot\sigma} = \mathbb{1}\cos\theta - i(\hat{n}\cdot\sigma)\sin\theta.$$

c) Check that the spinor $|\hat{n}; +\rangle$ is an eigenspinor of $\hat{n}\cdot\sigma$, which shows it is indeed an eigenspinor of $S_n = \hat{n}\cdot\mathbf{S}$.

Hints: Use the following three facts you will prove on the review: If H is hermitian, then e^{iH} is

unitary. If U_1, U_2 are two unitary operators, then $U_1 U_2$ is also unitary. If $U = e^{-\frac{i}{2}\alpha\sigma_z} e^{-\frac{i}{2}\beta\sigma_y} e^{-\frac{i}{2}\gamma\sigma_z}$, then

$$U^\dagger \boldsymbol{\sigma} U = R \boldsymbol{\sigma},$$

where R is the matrix which rotates a 3-vector first by α around $\hat{z}$, then by β around $\hat{y}$, and then by γ around $\hat{z}$ again. Also note that if $\mathbf{a}, \mathbf{b}$ are 3-vectors, then $\mathbf{a} \cdot R\mathbf{b} = R^{-1}\mathbf{a} \cdot \mathbf{b}$.

d) Repeat (b) and (c) for $|\hat{n}; -\rangle$.

a) Convinced! The explanation is that if $\hat{n}$ has polar angle β and azimuthal angle α , then we can first get rid of the polar angle by rotating around $\hat{y}$ and then get rid of the azimuthal angle by rotating around $\hat{z}$.

b) First, we need a simpler form for $e^{-\frac{i}{2}\beta\sigma_y}$. Using the formula in the hint, we find

$$e^{-\frac{i}{2}\beta\sigma_y} = \cos(\beta/2) - i \sin(\beta/2)\sigma_y.$$

On the other hand, σ_z is diagonal, so that

$$e^{-\frac{i}{2}\alpha\sigma_z} = \begin{bmatrix} e^{-\frac{i}{2}\alpha} & 0 \\ 0 & e^{\frac{i}{2}\alpha} \end{bmatrix}.$$

If you haven't seen this before, here's the proof: Let $D = \text{diag}(d_1, \dots, d_m)$ be a diagonal $(m \times m)$ -matrix. Then

$$e^D = \sum \frac{D^n}{n!} = \sum \frac{1}{n!} \begin{bmatrix} d_1^n & 0 & \cdots & 0 \\ 0 & d_2^n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & d_m^n \end{bmatrix} = \text{diag}(e^{d_1}, \dots, e^{d_m}).$$

Thus,

$$e^{-\frac{i}{2}\alpha\sigma_z} e^{-\frac{i}{2}\beta\sigma_y} = \begin{bmatrix} e^{-\frac{i}{2}\alpha} & 0 \\ 0 & e^{\frac{i}{2}\alpha} \end{bmatrix} \cos(\beta/2) - i \sin(\beta/2) \begin{bmatrix} 0 & -ie^{-\frac{i}{2}\alpha} \\ ie^{\frac{i}{2}\alpha} & 0 \end{bmatrix}.$$

We can then compute

$$|\hat{n}; +\rangle = e^{-\frac{i}{2}\alpha} \cos(\beta/2) |+\rangle + e^{\frac{i}{2}\alpha} \sin(\beta/2) |-\rangle.$$

c) Denote $|\hat{n}; +\rangle = U |+\rangle$, where U is a unitary operator by the facts in the hint. Then,

$$\begin{aligned} \hat{n} \cdot \boldsymbol{\sigma} |\hat{n}; +\rangle &= \hat{n} \cdot \boldsymbol{\sigma} U |+\rangle = U U^\dagger \hat{n} \cdot \boldsymbol{\sigma} U |+\rangle = \\ &= U \hat{n} \cdot (R \boldsymbol{\sigma}) |+\rangle = U (R^{-1} \hat{n}) \cdot \boldsymbol{\sigma} |+\rangle = U \sigma_z |+\rangle = \frac{\hbar}{2} U |+\rangle. \end{aligned}$$

d) This is the same calculation for (c). For (b), we end up with

$$|\hat{n}; -\rangle = e^{\frac{i}{2}\alpha} \cos(\beta/2) |-\rangle - i e^{-\frac{i}{2}\alpha} \sin(\beta/2) |+\rangle.$$